

Manufacturer Name: Well-Tanker (Anhui) M&E Equipment Co.,Ltd.

Address:Wushan Town Hefei City Anhui Province China

Description of Pressure Equipment: Name: Oil/Gas Separator Dwg no.: 20980145-083-001 Rev.3



SIMPLIFIED RISK ANALYSIS

1 - Variability of the stresses around the set point. Fatigue	Negligible	Moderate	Important	V. Important	Score
Status		X			
Risk Value	1	2	3	4	2
<i>Variability (amplitude and number of cycles) stresses due to pressure, temperature, vibration ... may involve, when it is important that there be a fatigue analysis. The realization of such an analysis does not relieve the need to take into account the variability of the stresses in the material selection, assembly design, the extent and severity of controls. It is particularly important to take into account the particularly harmful effects that may have some corrosion mechanisms on the fatigue behavior of the equipments.</i>					
2 - Frequency of starts and stops.	Negligible	Moderate	Important	V. Important	
Status		X			
Risk Value	1	2	3	4	2
<i>The starting frequency depends on the conditions of machine operation (operating intermittently, maintenance outages, catalyst regeneration or reagent product, for example ...). Cycles start and stop must be taken into account here regardless of the variability of stress.</i>					
3 - Very sharp variation of temperature or pressure	Negligible	Moderate	Important	V. Important	
Status	X				
Risk Value	1	2	3	4	1
<i>Conditions of use of a equipment include the nature and control of changes undergone by the fluid or fluids contained therein (change of physical state, chemical reactions, brutality of changing pressures and temperatures, unstable fluids. ...).</i>					
4 - Negative pressure inside or outside pressure.	Negligible	Moderate	Important	V. Important	
Status	X				
Risk Value	1	2	3	4	1
5 - The monitoring of the equipment in use	Continue	Periodic & s	Occasional	Nonexistent or Impossible	
Status	X				
Risk Value	1	2	3	4	1
<i>The monitoring equipment can be driven either by the operating personnel, or by control equipments and automatic control, or a combination of both. It is the subject of instructions called "operating instructions".</i>					

6 - In service inspection.						
Status						
Risk Value		X	1	2	3	4
The inspection in service is to examine or measure continuously or periodically if some features may change and possibly to demonstrate the occurrence of damage.						
7 - Complexity of the equipment.						
Status	X					
Risk Value		1	2	3	4	1
The complexity of the equipment due to its design. It concerns the general forms of the equipment and its internal and external equipment. The complexity of form is characterized by the presence of discontinuity, asymmetries ...						
8 - Possibility of degradation due to corrosion or erosion						
Status						
Risk Value		1	2	3	4	2
The possibility of damage due to corrosion, erosion and, more generally, to the deterioration of the material by the nature of fluids in contact with the wall.						
9 - The possibility of degradation in service related to						
Status	X					
Risk Value		1	2	3	4	1
The possibility of degradation due to creep rupture or brittle fracture.						
10 - Temperature of the product, in case of leakage, danger to the material						
Status	X					
Risk Value		1	2	3	4	1
These are physical risks associated with the fluid temperature (steam, hot gas, liquefied gas).						
11 - The population affected in case of failure.						
Status	X					
Risk Value		1	2	3	4	1
The affected population includes anyone who might be in the area affected by a possible failure of the equipment.						
12 - The presence of operating personnel in the vicinity of the equipment.						
Status						
Risk Value		1	2	3	4	2
The importance of staff presence due, firstly, the implantation of the equipment in the system considered and, secondly, the operating conditions of the unit.						

13 - The economic impact of failure.						
Status		Small	Moderate	Important	V. Important	
	X					
Risk Value		1	2	3	4	1
<i>The economic impact includes in particular: - the loss of production by immobilization of the facility - the overall cost of repair or replacement.</i>						
14 - The failure of the equipment it may cause failure of a neighbor equipment whose consequences would be?						
Status		Small	Moderate	Important	V. Important	
	X					
Risk Value		1	2	3	4	1
<i>The consequences of equipment failure are nearby to be evaluated.</i>						
Total Risk						19
Risk Value associated to PED RISK CATEGORY						
Art. 4.3		Status	Risk Value			
			0			
Category I			4			
Category II			8			
Category III	X		12			
Category IV			16			
PED Risk						12

Comprehensive evaluation of the potential factors of failure and the consequences of a possible failure	
Low	14-31
Moderate	32-45
High	46-59
Very High	60-72

Comprehensive Risk: 31
Risk Evaluation: LOW

Risk Level	Mesures suggested in order to reduce the				
	Efficiency	Allowable stress	Inspection Scope	Pressure test	PT Value
Low	0.7	10% Reduced	0	Only Hydro Test	Improved
Moderate	0.85	Nominal	10%	Pneumatic or HydroTest	Nominal
High	0.85	Nominal	25%	Pneumatic or HydroTest	Nominal
Very High	1	Nominal	100%	Hydro Test and further Test	Nominal

* These measures are only recommended. The requirements of the directive and standards adopted remain mandatory. If more stringent provisions are required by the code,